

Eliseu Venites Filho

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EDUCATION

Universidade Federal do Rio Grande do Sul

Porto Alegre - Brazil

Ph.D. in Physics (Computational Statistical Physics)

Jul. 2021 - Present

- Analysis of the ensemble correlations of observables of complex systems in order to predict their critical behavior
- Systems from different Universality Classes considered
- Both systems with and without a defined Hamiltonian considered
- The simulations were implemented in Rust while the data analysis was done in the Julia ecosystem

Universidade Federal do Rio Grande do Sul

Porto Alegre - Brazil

M.Sc. in Physics (Computational Statistical Physics)

Mar. 2019 - May 2021

- Scored higher than 99.42% of candidates on the EUF 2-2018 (National graduate programs entrance exam)
- Performance evaluation of the Simulated Annealing applied to different configurations of the Traveling Salesman Problem
- Analysis of the stochastic optimization algorithm applied to problems at the boundary between P and NP complexity classes
- The optimization algorithm was implemented in C++ while the data analysis was done in the Python ecosystem

Institut d'Optique Graduate School

Palaiseau - France

M.Sc. in Optical Engineering

Sep. 2015 - Sep. 2017

- Double degree in the context of BRAFITEC program
- Courses included: Optical Instrumentation, Automation, Lasers and Quantum Optics

Université ParisSud

Orsay - France

L3 and M1 in Fundamental Physics

Sep. 2015 - Sep. 2017

- Double degree magistère offered to engineering students
- Courses included: Analytical Mechanics, Statistical Physics, Plasma Physics and Atomic and Molecular Physics

Universidade Federal do Rio Grande do Sul

Porto Alegre - Brazil

B.Sc. in Engineering Physics

Mar. 2013 - Dec. 2018

- Scientific Initiation (CAPES) on Quantum Information in 2013 and 2014
- Presentation at the UFRGS XXVI Scientific Initiation Meeting (2014): Shor's Algorithm for Integer Factorization
- Summa Cum Laude with final grade 9.54/10.0

EXPERIENCE

Optical Engineering Internship

Paris, France

Télécom ParisTech

Jun. 2017 - Sep. 2017

- As part of the "Information Quantique et Applications" research group
- Worked with polarization-entangled photon pairs source
- Stabilization and count optimization of the entangled photon pair source to be used in experiments testing Quantum Key Distribution protocols

PUBLICATIONS

[Revisiting the Contact Model with Diffusion Beyond the Conventional Methods](#)

2025

Symmetry

R. da Silva, E. Venites Filho, H. A. Fernandes, P. F. Gomes

[Efficient computational method using random matrices describing critical thermodynamics](#)

2025

International Journal of Modern Physics C

R. da Silva, E. Venites Filho, S. D. Prado, J. R. D. de Felício

[A Spectral Investigation of Criticality and Crossover Effects in Two and Three Dimensions: Short Timescales with Small Systems in Minute Random Matrices](#)

2024

Entropy

E. Venites Filho, R. da Silva, J. R. Drugowich de Felício

[Mean-Field Criticality Explained by Random Matrices Theory](#)

2023

Brazilian Journal of Physics

R. da Silva, H. C. M. Fernandes, E. Venites Filho, S. D. Prado, J. R. Drugowich de Felício

[A Thorough Study of the Performance of Simulated Annealing in the Traveling Salesman Problem under Correlated and Long Tailed Spatial Scenarios](#)

2021

Physica A: Statistical Mechanics and its Applications

R. da Silva, E. Venites Filho, A. Alves

PROJECTS

[tsp-sa](#) | C++, Python

- Developed in the context of the M.Sc. research
- Modular C++ library to perform optimization through Simulated Annealing
- Supports Generalized Simulated Annealing and Tsallis Entropy statistics
- Optimization logic works for arbitrary Markov chains, completely decoupled from the TSP implementation
- Data analysis and plotting done in Python

[artificial-systems](#) | Rust (ndarray, serde)

- Developed in the context of the Ph.D. research
- Computational models of artificial systems implemented in Rust
- Simulation of Spin Systems (Ising and Blume-Capel models)
- Investigation of the Contact Process with diffusion

[ts-cov-matrix](#) | Julia (DataFrames.jl, Makie.jl)

- Developed in the context of the Ph.D. research
- Analysis of time series covariance matrices using Random Matrix Theory
- Study of spectral properties and comparison with Marchenko-Pastur distribution
- Analyzed data from NOAA temperature records, Spin Systems, and Contact Processes
- Full data analysis pipeline implemented in the Julia ecosystem

[json-parser](#) | Haskell

- Strict JSON parser implemented in Haskell using Megaparsec
- Adheres closely to JSON standards
- Can be used as a library or a standalone command-line tool

[sternhalma-server](#) | Rust (tokio)

- Asynchronous game server for Sternhalma (Chinese Checkers) built with Rust and Tokio
- Actor-like architecture with decoupled game logic and connection handling
- Client-agnostic design supporting CLI, GUI, and AI agents
- Supports both Raw TCP and WebSocket connections using a CBOR-based protocol

[sternhalma-agent \(WIP\)](#) | Python (PyTorch)

- Reinforcement learning agent implementing AlphaZero from scratch
- Uses Monte Carlo Tree Search (MCTS) for planning and Deep Neural Networks (ResNet) for evaluation
- Designed to master Sternhalma through self-play without human knowledge

SKILLS

Languages: Portuguese (native), English (fluent), French (advanced)

Programming: C, C++, Rust, Python, Julia, Haskell

Libraries: tokio, ndarray, serde, faer-rs, Pandas, NumPy, SciPy, scikit-learn, matplotlib, PyTorch, TensorFlow, DataFrames.jl, Plots.jl, Makie.jl

Tools: Linux, Git, Docker, SQL

Typesetting: LaTeX, Typst